University of Bristol — Data Science: Visual Analytics

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Housing Insecurity and Economic Integration of International Migrants in England & Wales (2011–2021)

A Visual Analytics Dashboard for Policy Decision Support

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Abstract

This report presents a five-tab Tableau dashboard analysing the housing insecurity and economic integration of international migrants across 331 Local Authority Districts (LADs) in England and Wales, using the 2011 and 2021 ONS Census. The dashboard is organised as a **cross-tab decision funnel** (Tab 1 → Tab 2 → Tab 5 → Tab 3 → Tab 6) implementing five decision scenarios for Home Office policy analysts and local integration officers: migrant-pressure screening, structural-precarity screening, forward-looking 2031 forecast, intervention-type classification, and peer benchmarking. Principal Component Analysis (PCA) with two components captures 85.7% of the variance across eight socioeconomic indicators, revealing a New Arrival & Language Barrier axis (PC1 = 61.2%) and a Welfare Dependency axis (PC2 = 24.5%). K-means with $k = 4$ (selected by Gaussian Mixture Model BIC minimum at 2463.8) yields four interpretable cluster types — Migration Pressure (41 LADs), Health Crisis (85), Watch Only (109), Healthy & Stable (96). Bayesian Ridge Regression projects social renting and economic inactivity rates to 2031 with 95% credible intervals, with seven LADs flagged at *Critical* composite risk (Hartlepool, Middlesbrough, Newcastle upon Tyne, South Tyneside, Sunderland, Liverpool, Nottingham) — all in the Health Crisis cluster, indicating that the highest 2031 risk is concentrated on welfare-dependency LADs rather than the high-migrant Migration Pressure cluster. UMAP-based within-cluster z-score analysis identifies positive benchmarks for case-study transfer (e.g. Wandsworth as a Dual benchmark for the Migration Pressure cluster, $z^{\text{combined}} = -2.57$; Barrow-in-Furness as the single Dual benchmark for the Health Crisis cluster, $z^{\text{combined}} = -2.11$). The dashboard demonstrates how probabilistic uncertainty quantification, cluster-aware peer benchmarking, and an explicit funnel contract between tabs can be embedded directly into policy visualisations to support well-tuned, checkable decision-making.

1 Introduction

1.1 Socio-Economic Context

International migration is among the most important and debated policy domains in contemporary Britain. Between the 2011 and 2021 censuses, the foreign-born share of England and Wales' population grew substantially (Office for National Statistics, 2022d), driven by a combination of EU free movement (pre-2020), post-Brexit managed migration, asylum seekers, and international students (Migration Observatory, 2022). Yet the distribution of migrants across local areas is deeply uneven. Tower Hamlets, Brent, and Haringey host foreign-born populations exceeding 35% of their residents, while rural authorities in the East Midlands record below 5% (Office for National Statistics, 2022b). This spatial concentration creates structural pressures on housing markets, labour market integration, and public service demand (Dustmann and Frattini, 2014) — pressures that fall unequally on LADs with limited financial independence.

The decade 2011–2021 spans two major structural breaks: the post-2016 Brexit referendum that reshaped EU migration flows (Wadsworth et al., 2016), and the 2020–2021 COVID-19 pandemic that shortened the final census window (Office for National Statistics, 2022c). A visualisation tool designed for this period must therefore distinguish between worsening caused by structural vulnerability and short-term pandemic effects.

1.2 Problem Statement

Policy intervention in migrant integration is complicated by three analytical challenges:

1. **Many dimensions**: Housing insecurity, economic inactivity, English language proficiency, and health outcomes co-vary in complex ways. No single indicator captures the full picture of integration difficulty.

2. **Geographic scale**: With 331 LADs, manual review is impractical. A visualisation must reduce the search space while preserving the ability to inspect individual cases.

3. **Temporal uncertainty**: Projecting needs to 2031 — the planning horizon for the UK's Integrated Settlements Strategy (Department for Levelling Up, Housing and Communities, 2022) — requires uncertainty quantification, not merely point estimates.

1.3 Target Users

The primary users of this dashboard are **Home Office policy analysts** responsible for allocating integration support budgets (English for Speakers of Other Languages [ESOL] programmes, housing support grants, employment linking services) across LADs (Ministry of Housing, Communities and Local Government, 2018; Home Office and Ministry of Housing, Communities and Local Government, 2019). Secondary users are **local government integration officers** who need to understand their own area's position relative to regional peers.

These users are data-literate but not specialist statisticians. They can interpret choropleth maps, scatter plots, and bar charts without guidance, but require explicit labelling to interpret dimensionality reduction outputs. They operate under time pressure and need to reach defensible, checkable conclusions. Their primary mental model is geographic ("which areas in the North West?") rather than abstract ("which clusters?").

1.4 Scope and Limitations

The census captures usual residents (those intending to stay 12 months or more) (Office for National Statistics, 2022a), meaning short-term visa holders and undocumented migrants are excluded from both 2011 and 2021 counts. Economic inactivity and social renting rates therefore under-count the most precarious recent arrivals. The 2021 census was conducted during COVID-19 restrictions, which may have suppressed response rates in high-migrant urban areas (Office for National Statistics, 2022c).

2 Data Preparation and Abstraction

2.1 Data Sources

Two census datasets were obtained from NOMIS (nomisweb.co.uk), the Office for National Statistics' data sharing platform:

Year	Tables Used	Topic	Format
2011	KS204EW, QS203EW, QS801EW, KS206EW, KS301EW, KS402EW, KS601EW, QS205EW	Country of birth, tenure, economic activity, household language, health	Apple Numbers (.numbers)
2021	TS004, TS012, TS015, TS025, TS037, TS054, TS066, TS029	Equivalent 2021 tables	CSV (NOMIS bulk download)

Table 1: Census data sources by year.

The 2011 files required manual export from Apple Numbers format to CSV before programmatic processing. Geography codes follow ONS LAD boundaries (format E06/E07/E08/E09xxxxx), with the 2021 series using updated LAD21 codes.

2.2 Geography Harmonisation

A critical data preparation challenge is the boundary mismatch between 2011 and 2021 census geographies. Between the two censuses, **31 LAD codes changed** through mergers and code reassignments—all changes were mergers (no splits occurred). These 31 2011 LADs combined into 14 new 2021 LAD boundaries, reducing the total LAD count from 348 to 331 (a net reduction of 17). Representative examples include:

- **4→1 merger**: Corby, Kettering, East Northamptonshire, and Wellingborough → North Northamptonshire (`E06000061`)
- **3→1 merger**: Bournemouth, Poole, and Christchurch → Bournemouth, Christchurch and Poole (`E06000058`)
- **5→1 merger**: North Dorset, Purbeck, West Dorset, East Dorset, and Weymouth and Portland → Dorset (`E06000059`)

The boundary lookup was sourced from the ONS Open Geography Portal (Office for National Statistics, 2021a), specifically the `LAD11_LAD21_EW_LU` Feature Service, accessed programmatically via the ArcGIS REST API (Esri, 2024) using page-by-page `GET` requests with `resultOffset` increments of 1,000 rows until `exceededTransferLimit` returned `false`. The retrieved records were saved to `lad_2011_to_2021_lookup.csv` (348 rows covering all England and Wales LADs).

Boundary change inspection. The downloaded lookup was inspected programmatically by filtering rows where `LAD11CD` $\neq$ `LAD21CD`, identifying **31 changed codes**. These were then grouped by target `LAD21CD` to reveal the merger structure:

- **5→1**: North Dorset, Purbeck, West Dorset, East Dorset, Weymouth and Portland → Dorset
- **4→1**: Aylesbury Vale, Chiltern, South Bucks, Wycombe → Buckinghamshire; and Corby, Kettering, East Northamptonshire, Wellingborough → North Northamptonshire
- **3→1**: Bournemouth, Poole, Christchurch → Bournemouth, Christchurch and Poole; and Daventry, Northampton, South Northamptonshire → West Northamptonshire
- **Code reassignments only** (name unchanged): Northumberland, Gateshead, St Albans, Welwyn Hatfield, and others

All changes were mergers; no LAD splits occurred between the two censuses. For merged LADs, raw counts from component 2011 areas were summed before percentage derivation, preserving the correct denominator. Without this boundary matching step, the 31 affected 2011 LADs would have been silently

dropped from the 2011–2021 comparison, introducing systematic bias against areas that underwent administrative restructuring.

After boundary matching, the final merged dataset contains **331 LADs** (all valid LADs in England and Wales).

Geography hierarchy. ONS organises sub-national statistics across three nested levels relevant to this dashboard. A **geography code** (also called LAD code) is the primary unit of analysis: a 9-character ONS identifier (format `E06/E07/E08/E09xxxxxx` for England; `W06xxxxxx` for Wales) denoting a single Local Authority District. Each LAD is then nested within a **region**: one of nine English administrative regions (e.g. London, North West, Yorkshire and The Humber) or Wales as a single unit. Regions aggregate multiple LADs and serve as the categorical variable used for colour-coding throughout the dashboard.

Region lookup acquisition. To attach region names to each LAD, two additional lookup tables were downloaded from the ONS Open Geography Portal via the ArcGIS REST API using the same page-by-page `GET` method described above:

- `lad21_to_region.csv`: sourced from the `LAD21_RGN21_EN_LU` Feature Service (Office for National Statistics, 2021b), fields `LAD21CD` and `RGN21NM`, covering all 331 England LADs. Wales LADs (prefix `W06`) are not included in this England-only service and are assigned the label "Wales" by prefix detection in code.
- `lad11_to_region.csv`: sourced from the `LSOA11_BUASD11_BUA11_LAD11_RGN11_EW_LU` Feature Service (Office for National Statistics, 2011). Because this service is published at LSOA grain (multiple rows per LAD), records were deduplicated on `LAD11CD` before saving, yielding one row per 2011 LAD.

In practice, only `lad21_to_region.csv` is consumed at runtime. Because all 2011 LAD codes are recoded to their 2021 equivalents *before* the region join (see above), every `geography_code` key at the point of lookup is already a LAD21CD. The 2011 region file was downloaded for completeness but is not required by the analysis pipeline.

2.3 Variable Derivation

All raw count variables were normalised to percentages using the appropriate population denominator, enabling comparison across LADs of varying sizes:

Derived Column	Formula	Denominator
`migrant_pct`	(total_pop − uk_born) / total_pop × 100	Total resident population
`social_rent_pct`	social_rented_hh / total_hh × 100	Total households
`econ_inactive_pct`	economically_inactive / pop_16plus × 100	Population aged 16+
`no_english_pct`	no_english_hh / total_hh × 100	Total households
`bad_health_pct`	(bad_health + very_bad_health) / total_pop × 100	Total resident population
`disability_pct`	disabled_under_ea / total_pop × 100	Total resident population
`eu_born_pct`	eu_born / total_pop × 100	Total resident population
`post_transition_arrival_pct`	arrived_2020_2021 / total_foreign_born × 100	Total foreign-born residents

Table 2: Derived percentage variables and their denominators.

Change variables were derived by subtraction (e.g., `social_rent_change` = `social_rent_pct_2021` − `social_rent_pct_2011`), preserving signed values to distinguish improvement from worsening.

2.4 Missing Values and Data Quality

The downloaded NOMIS census tables provide complete counts at local authority level — ONS publishes figures for all geographic units without suppression at this geographic scale. As a result, **no missing values** were present in the raw count columns after the geography boundary matching step.

However, the 31 boundary changes affected **31 2011 LADs** that were merged into 14 2021 boundaries. For these LADs, 2011 counts from component areas were summed to match the 2021 boundary before percentage derivation. No proportional splitting was required, as all changes were mergers. The Bayesian forecast model (Section 4) indirectly takes in remaining boundary uncertainty through the decade-scale residual standard deviation term added to the credible interval.

For the Bayesian Ridge forecast (Tab 5), an `IterativeImputer(estimator=BayesianRidge())` was applied as a precautionary step to handle any residual NaN values following boundary matching. In practice, no imputation was required, confirming the completeness of the final dataset.

2.5 Data Abstraction

Following Munzner (2014) (Ch. 2), the raw tables were transformed into a domain-independent data abstraction:

Data type	Representation	Example
Items	Local Authority Districts (LADs)	Hackney, Leeds
Attributes (quantitative)	Percentage indicators (8 per LAD per year)	social_rent_pct_2021 = 23.4%
Attributes (categorical)	Region, cluster_name, above_threshold	London, Health Crisis, Certain crisis
Attributes (ordered sequential)	Year (2011, 2021), forecast year (2031)	year = 2021
Derived attributes	Change deltas, PCA coordinates, UMAP coordinates, forecast + CI	social_rent_change = +2.1pp
Spatial data	LAD polygon boundaries (2021 LAD shapefile, BFE variant)	E09000012 Hackney

Table 3: Data abstraction following Munzner (2014).

The principal data type is **tabular** (one row per LAD, or one row per LAD per year for the trajectory trend chart). PCA and UMAP projections transform the 8-dimensional quantitative space into a 2-dimensional spatial representation. The Bayesian forecast adds two probabilistic attributes (credible intervals for social rent and economic inactivity) that are not directly observable from the census but derived from the temporal trend.

3 Task Definition

Following Munzner's three-level task abstraction (Munzner, 2014) (Ch. 3) — Analyze (high), Search (mid), Query (low) — the dashboard is structured as a **cross-tab decision funnel**: each tab applies a narrowing rule, a diagnostic label, or a peer-transfer transformation to the LAD pool, and passes a defined output (`geography_code` or `cluster_name`) to the next tab. Five decision scenarios are implemented across the funnel (Tab 1 → Tab 2 → Tab 5 → Tab 3 → Tab 6), ending in a single concrete policy action: a peer LAD plus 1–2 indicators on which to commission a policy-transfer case study.

Tab	Funnel role	Decision scenario	Output to next tab
1	Narrower (1/3)	Decision 1-1 — Migrant pressure screening	∼30–50 LADs (Q1 of 1-3-b: $\Delta\text{share} > 0 \wedge \Delta\text{count} > 0$)
2	Narrower (2/3)	Decision 1-2 — Structural precarity screening	64 Q1 LADs (SR $\geq 15.85\% \wedge$ EI $\geq 39.51\%$) $\supset$ 6 emergency
5	Narrower (3/3)	Decision 2 — Forward-looking forecast	∼10–30 LADs (`composite_risk_tier` ∈ {Critical, High Risk})
3	Labeller	Decision 3 — Intervention classification	Priority LADs each tagged with one of four `cluster_name`s
6	Transformer (final)	Decision 4 — Peer benchmarking	1 peer LAD + 1–2 indicators per priority LAD (final — no further handoff)

Table 4: Cross-tab decision funnel. Each row gives the tab number, its funnel role (narrower / labeller / transformer), the decision scenario it implements, and the output it passes to the next tab. The complete handoff contract is documented in `notebooks/tab_prior_target_LAD_funnel_process.md`.

3.1 Decision 1-1 — Migrant Pressure Screening: "Where are migrants concentrated and growing?"

Policy decision: Home Office integration analyst identifies the LADs where migrant share *and* migrant headcount both grew between 2011 and 2021 — i.e., areas under genuine and sustained integration pressure rather than mere demographic noise.

Munzner Level	Value
Analyze	Consume → **Discover** (which areas grew on both axes is not known in advance)
Search	**Explore** (neither target LAD nor location is known)
Query	**Summarize** (overview map first) → **Compare** (sorted bar chart) → **Identify** (Q1 of 1-3-b growth quadrant)
Target	Attributes → *Distribution* (spatial spread) + *Extremes* (top-share LADs) + *Trends* (2011→2021 change)
Derived attribute	`migrant_pct`, `migrant_pct_change`, `migrant_cnt_change` (computed via FIXED LOD per `geography_code`)

Dashboard location: Tab 1 (Sheets 1-1 symbol map, 1-2 ranked bar, 1-3 change choropleth, 1-3-b growth quadrant scatter).

A **chained task sequence** is implemented: Sheet 1-1 (where in 2021) → 1-2 (rank by share) → 1-3 (where the share grew) → 1-3-b (where both share *and* count grew = Q1 prior target). The decision rule is the upper-right quadrant of 1-3-b: `migrant_pct_change` > 0 AND `migrant_cnt_change` > 0.

3.2 Decision 1-2 — Structural Precarity Screening: "Which of those areas are also under structural precarity?"

Policy decision: Of the LADs surfaced by Decision 1-1, identify those that also clear the dual structural-precarity threshold (high social rent + high economic inactivity), narrowing to the 64-LAD Q1 pool and the ∼6-LAD emergency shortlist within it.

Munzner Level	Value
Analyze	Consume → **Discover** (which LADs are jointly above both thresholds is not known)
Search	**Locate** (find specific LADs in the upper-right quadrant)
Query	**Identify** (Q1 quadrant membership) + **Compare** (per-LAD position relative to the two threshold lines)
Target	Attributes → *Correlation* (joint distribution of SR and EI) + *Extremes* (Tukey-fence emergency outliers labelled in-chart)
Derived attribute	`social_rent_pct_2021`, `econ_inactive_pct_2021`, `quadrant` (4-level categorical), `is_q1_prior_target` (boolean)

Dashboard location: Tab 2 (Sheet 2-1 quadrant scatter + Sheet 2-2 spatial continuity map). Decision rule: Q1 = SR $\geq 15.85\%$ AND EI $\geq 39.51\%$ (the national means). Setting the "Select Q1 Prior Targets" filter to True labels the emergency shortlist (Nottingham, Newcastle upon Tyne, Blaenau Gwent, Sunderland, South Tyneside, Liverpool).

3.3 Decision 2 — Forward-Looking Forecast: "Where will housing and economic insecurity peak by 2031?"

Policy decision: DLUHC (Department for Levelling Up, Housing and Communities) selects LADs for the 10-year integration strategy *before* crisis becomes visible in current data, ranking by the 2031 composite risk tier (housing + economic forecast).

Munzner Level	Value
Analyze	Consume → **Present** (communicate probabilistic risk to decision-makers who must act now)
Search	**Browse** (scan trajectory lines for rising slopes; scan map for hot regions)
Query	**Identify** (which LADs have CI lower exceeding the threshold — *certain* crisis?) + **Compare** (which worsen on both housing *and* economic dimensions?)
Target	Attributes → *Extremes* (LADs with highest forecasts) + *Trends* (2011→2031 trajectory in Sheets 5-1 / 5-1-b)
Derived attribute	`forecast_social_rent_2031`, `forecast_econ_inactive_2031`, `*_ci_lower/upper` (BayesianRidge posterior predictive); `housing_risk_tier` / `economic_risk_tier` / `composite_risk_tier` (5-level percentile classification)

Dashboard location: Tab 5 (Sheets 5-1 / 5-1-b trajectory line charts, 5-2 composite-risk choropleth, 5-3 priority quadrant scatter). Decision rule: `composite_risk_tier` ∈ {Critical, High Risk}.

Tab 2 defines insecurity as the intersection of high social renting *and* high economic inactivity. Tab 5 maintains this dual-metric framework by forecasting both dimensions independently using BayesianRidge, trained on 2011→2021 data and projected forward to 2031. The trajectory line charts (Sheets 5-1 and 5-1-b) display the full 2011→2021→2031 arc for each LAD as connected solid (actual) and dashed (forecast) line segments, embedding the temporal-change view directly within the forecast tab rather than splitting it into a separate descriptive trajectory tab. The key contribution of the Bayesian approach is distinguishing *certain* from *likely* risk through the credible interval: an area where the lower CI already exceeds the risk threshold will worsen even under the optimistic scenario.

3.4 Decision 3 — Intervention Typology: "What kind of problem does each LAD have?"

Policy decision: For each priority LAD surfaced by Decisions 1-1, 1-2, and 2, local integration officers select the intervention type (ESOL classes, housing support, employment linking, NHS / disability-welfare resourcing, or a mixed programme) by reading off the LAD's diagnostic `cluster_name` (Home Office and Ministry of Housing, Communities and Local Government, 2019).

Munzner Level	Value
Analyze	Consume → **Discover** (cluster structure is unknown) + Produce → **Annotate** (label every LAD with its cluster)
Search	**Explore** (open-ended: does a classification of LADs exist?) + **Lookup** (read off cluster for a known priority LAD)
Query	**Summarize** (overview of cluster layout) → **Identify** (click a point → which cluster, which features?)
Target	Attributes → *Similarity* (8-dimensional similarity reduced to 2D) + *Features* (cluster z-score signatures as policy prototypes)
Derived attribute	`pca_x`, `pca_y` (PC1, PC2 coordinates); `cluster_name` (4 policy-interpretable labels); `kmeans_assignment_prob`, `kmeans_gmm_agree` (uncertainty signals)

Dashboard location: Tab 3 (Sheets 3-1 PCA scatter with K-means + GMM, 3-2 PC1 loading bar, 3-3 PC2 loading bar, 3-4 cluster z-score heatmap). This tab is a **labeller**, not a narrower: pool size is unchanged but every LAD now carries a `cluster_name` that joins on `geography_code` into Tab 6. The PCA scatter is paired with two loading bar charts (PC1, PC2) and a cluster z-score heatmap, so the user can answer *why* an area sits where it sits in the scatter — a "details on demand" interaction (Shneiderman, 1996). This directly supports the intervention-type decision.

3.5 Decision 4 — Peer Benchmarking: "Who succeeded with similar conditions?"

Policy decision: For each priority LAD (now carrying a `cluster_name`), the policy-evaluation team identifies one peer LAD in the *same cluster* that clearly outperforms its peers, and names the 1–2 indicators on which the gap is largest. The output is a policy-transfer case-study brief, not another filter.

Munzner Level	Value
Analyze	Consume → **Discover** (positive outlier LADs are unknown in advance) + Produce → **Derive** (recommend a peer for case-study transfer)
Search	**Explore** (looking for anomalously well-performing LADs within cluster)
Query	**Compare** (outlier LAD vs. cluster median; per-feature gap bar chart)
Target	All Data → *Outliers* (LADs anomalously better than their cluster peers on social rent and/or economic inactivity)
Derived attribute	`umap_x/y`; within-cluster z-scores z_{sr}, z_{ei} on the two topic axes; composite `outlier_type` ∈ {Dual, Housing only, Economic only, Peer-aligned}; `benchmark_strength` ($= -z^{\text{combined}}$); per-feature gap vs. cluster median (Sheet 6-3)

Dashboard location: Tab 6 (Sheet 6-1 UMAP scatter, 6-2 outlier-type profile heatmap, 6-3 per-LAD gap bar chart, 6-4 ranked list of top benchmarks). This tab is a **transformer**, not a filter: it takes a struggling priority LAD as input and returns a *different* LAD (a peer outlier in the same cluster) plus the indicator names justifying the policy-transfer case study.

4 Visualisation Justification

4.1 Design Principles

The dashboard was designed with three cross-cutting principles grounded in perception and cognition research:

1. **Accurate channels for primary tasks**: For ranking tasks, position along a common scale is used (bar charts) — the most accurate visual channel (Cleveland and McGill, 1984). For similarity tasks, spatial position in a 2D scatter is used (PCA/UMAP). For change-over-time, connected line charts with distinct actual/forecast segments show the full 2011→2021→2031 trajectory within a single view.

2. **Colour for categorical discrimination, not decoration**: Hue is used only for unordered categorical data (region, cluster type, period); sequential colour scales are used for ordered quantitative data; diverging scales are used only where a meaningful zero exists (change values, PCA loadings). This follows Bertin's (1983) retinal variable hierarchy and Munzner's (2014) channel effectiveness principles.

3. **Overview then detail** (Shneiderman, 1996): Every tab opens with a full-dataset view (choropleth, scatter, bar chart), with detail accessible on demand via tooltip or Dashboard Action. This reduces cognitive load for users who only need to confirm a prior hypothesis, while supporting open-ended exploration for users who do not.

4.2 Tab 1 — Migration Overview: Geographic Distribution and Change

Tab 1 comprises four sheets that together address Decision 1 (Geographic Screening). Each sheet implements a distinct policy question; together they form a complete information architecture for screening.

Sheet 1 — Choropleth Map (migrant_pct, stock view)

Idiom choice: A filled choropleth map is the standard idiom for an area-level attribute distributed over geography. Spatial position directly encodes the most natural "where" question for policy users whose mental model is geographic (Shneiderman, 1996). The colour attribute (migrant percentage) uses a **sequential single-hue scale (White → Orange-Red)**, appropriate for an ordered quantitative variable with a natural zero. A diverging scheme would imply a meaningful midpoint, which does not exist here.

EDA validation: The distribution of `migrant_pct` is right-skewed (mean 14.2%, median 10.4%, max 56.1%). This skew justifies the sequential palette choice: a diverging palette would impose a false midpoint on a one-sided distribution.

Animation (Pages: Year): Displaying 2011 and 2021 as a flip-book animation exploits automatic motion perception (Ware, 2004). Temporal change that would require side-by-side comparison in a

static view is perceived immediately through motion. EDA confirmed that over 80% of LADs lie above the no-change diagonal (i.e., migrant share increased 2011–2021), making the animation substantively meaningful rather than cosmetic.

Region filter: EDA reveals that the London median migrant percentage (c. 40%) is more than three times the national median (c. 10%). Without a region filter, London LADs would dominate every ranking and obscure policy-relevant variation in other regions. The filter is implemented as a multi-value dropdown linked across all four sheets.

Sheet 2 — Top 20 Bar Chart (migrant_pct, stock ranking)

Limitation of choropleth: Choropleth maps conflate large, low-density areas visually (England's rural LADs cover more pixel area than London's dense ones). The companion bar chart corrects this by encoding migrant percentage via **position along a common baseline** — the perceptually most accurate channel for quantitative comparison (Cleveland and McGill, 1984).

This bar chart answers: *"Which areas currently have the highest migrant concentration?"* It is linked to Sheet 1 via a Dashboard Action (selecting a bar highlights the corresponding LAD on the choropleth).

Sheet 2b — Top 20 Bar Chart (migrant_cnt_change, flow ranking)

Important: Sheet 2 (stock) and Sheet 2b (flow) answer different policy questions and must not be conflated. Sheet 2 ranks areas by current migrant share; Sheet 2b ranks by *absolute increase in migrant headcount 2011–2021*. Service demand (ESOL class capacity, housing support caseloads) is proportional to the number of people, not to their share of the local population. A large urban LAD with a moderate migrant share may have received more new migrants in absolute terms than a small LAD with a higher share.

Important: `migrant_cnt_change` is stored in the CSV as a single per-LAD value, replicated across both year rows for compatibility with Tableau's long-format data model. In Tableau, this field must be aggregated with `MAX()` (not `SUM()`), since summing both year rows would double-count the change.

Idiom: Horizontal bar chart (identical idiom to Sheet 2) sorted by `MAX(Migrant Cnt Change)` descending, with a reference line at zero. The identical idiom across Sheets 2 and 2b makes the charts directly comparable and reduces the user's learning load. Colour encodes region (categorical hue).

Sheet 3 — Change Map (migrant_pct_change, integration pressure)

Important: `migrant_pct_change` is a **percentage point difference** in migrant share ($\texttt{migrant_pct}_{2021} - \texttt{migrant_pct}_{2011}$), not a relative change in migrant count. It measures *integration pressure*: how much larger the migrant share of the local population has become, regardless of whether total population also changed. An area where total population shrank could show a positive `migrant_pct_change` even with an unchanged migrant headcount. For this reason, the Sheet 3 tooltip includes `MAX(Migrant Cnt Change)` as a secondary indicator to flag such cases.

Colour palette: A **Purple–White–Orange (PuOr) diverging scale** is used, with zero centred at white. This palette was selected over Red–White–Blue to avoid UK political colour associations. Both ends are perceptually equal in intensity (Harrower and Brewer, 2003), satisfying Munzner's (2014) requirement that diverging scales be symmetric around the neutral midpoint.

LOD expression: Because the change map must display a single derived value per LAD without a Pages shelf, the change is computed via a Tableau FIXED LOD expression:

```
{ FIXED [Geography Code] : MAX(IF [Year]=2021 THEN [Migrant Pct] END) } − { FIXED [Geography Code] : MAX
```

This avoids the need for a separate wide-format data source and is recomputed consistently regardless of any downstream view filter.

4.3 Tab 2 — Scatter Plot: Encoding Multivariable Risk

Idiom choice: A scatter plot with two quantitative axes is the standard idiom for displaying the **correlation structure** between two continuous attributes (Anscombe, 1973). It allows simultaneous

encoding of four data dimensions: position-X (social rent %), position-Y (economic inactivity %), colour (region, 9 categories), and size (migrant %).

Four-quadrant reference lines: National average reference lines (Analytics → Average Line) divide the scatter into a 2×2 decision matrix. This exploits Gestalt region grouping (perceived without effort) (Ware, 2004): the user classifies any LAD by its quadrant without reading numbers. The top-right quadrant — "High housing stress + High economic inactivity" — is annotated as the priority intervention zone.

Extreme outlier labelling (Tukey IQR fence): Within the upper-right quadrant ($n = 64$ LADs), a second-tier filter identifies the most urgent cases for immediate budget allocation. The threshold follows Tukey (1977)'s standard upper fence: $Q_3 + 1.5 \times \text{IQR}$, applied independently to each axis within the quadrant. This is the conventional, citation-backed definition of a potential outlier — not an empirically chosen percentile. LADs exceeding the fence on either axis are labelled directly on the scatter, providing an automatic "pop-out" signal (Treisman and Gelade, 1980) that directs attention to the most acute cases without requiring the user to scan all 64 quadrant points.

Colour channel (region): Categorical hue is the natural channel for unordered nominal data (Bertin, 1983). Nine regions approach the practical distinguishability limit (Ware, 2004), but the categories are familiar to the user population, reducing the cognitive load of legend reading.

Size channel (migrant %): Size encodes a third contextual variable. Following Cleveland and McGill (1984), size (area) is less accurate than position, making it appropriate for a variable that contextualises the primary comparison but is not the primary analysis axis.

Colourblind safety: The 9-region palette avoids simultaneous red-green adjacency. The Tableau "Color Blind" preset is applied.

4.4 Tab 3 — PCA Scatter: Encoding Multivariate Similarity

Why PCA for Intervention Type Classification

PCA was selected over UMAP for Tab 3 for one critical reason: **interpretable axes**. The loading bar chart (Sheet 2) maps the original 8 indicator variables onto PC1 and PC2, enabling a policy user to explain *why* an area is positioned where it is in the scatter.

From the fitted PCA:

- **PC1 (X axis) = New Arrival & Language Barrier axis** (61.2% variance explained): Top positive loadings are Born outside UK (+0.438), Arrived 2011–21 (+0.432), No English in HH (+0.417). All three capture the same process: areas with large migrant stocks, recently arrived populations, and households without English — the integration challenge axis. Areas to the right have high migrant shares and language barriers; areas to the left have low migrant concentration but higher disability and economic inactivity rates.
- **PC2 (Y axis) = Welfare Dependency axis** (24.5% variance explained): Top positive loadings are Bad/very bad health (+0.582, NHS dependency), Social rented (+0.435, social housing dependency), Disabled (+0.432, disability benefit dependency), and Economically inactive (+0.342, out-of-work welfare dependency). All four components represent populations dependent on state-provided services. Areas at the top are high across all four welfare dependency dimensions.
- **Total variance captured: 85.7%** — justifying 2-component projection as representing almost all meaningful structure in the 8-feature space.

With UMAP, an equivalent axis interpretation is impossible. A policy user examining a specific LAD's placement could not identify which indicators placed it there, making that technique unsuitable for the intervention-type decision scenario.

GMM Soft Clustering for Assignment Uncertainty

A Gaussian Mixture Model scatter (Sheet 2, same PCA coordinate space) is displayed alongside the K-means PCA scatter. While K-means assigns each LAD to exactly one cluster using Euclidean distance to centroids, GMM models each cluster as a *multivariate Gaussian* with its own covariance structure and returns a *probability* of membership for each of the four components (**?**). K-means and GMM are run on the same 2D PCA coordinates, so the two scatters are spatially comparable.

GMM component naming via nearest K-means centroid: GMM components are labelled by the K-means cluster name whose centroid is nearest in PCA space. This is a naming convention to maintain visual consistency across the two views; it is *not* a claim that K-means and GMM produce identical partitions. The two algorithms partition the data differently: K-means yields a 96 / 109 / 41 / 85 split, whereas GMM yields a 103 / 143 / 39 / 46 split. In particular, the GMM component labelled *Watch Only* (143 LADs) is broader than its K-means counterpart (109 LADs) and takes in LADs that K-means assigns to *Health Crisis*, because GMM's covariance-aware (Mahalanobis-like) distance fits an elongated Gaussian over the low-PC1 region while K-means' Euclidean cut splits it more evenly along PC2.

Two complementary uncertainty signals: With this naming alignment, two distinct uncertainty signals are exported:

- `max_membership_prob` — the maximum of the four GMM probabilities for an LAD; this is GMM's confidence in *its own* most-likely component. Useful for the GMM scatter (Sheet 2) where colour reflects GMM's own argmax.
- `kmeans_assignment_prob` — the GMM probability for the component whose name matches the LAD's K-means cluster. This answers *"how confident is GMM in the K-means assignment?"* and is the right size encoding when colour is K-means `cluster_name` (the primary scatter, Sheet 1).
- `kmeans_gmm_agree` — a boolean flag capturing whether the LAD's K-means and GMM cluster names match (i.e., whether the GMM argmax falls in the component nearest to the K-means cluster). True for 243 / 331 LADs (73.4%).

These three signals decompose uncertainty along two orthogonal axes (within-GMM ambiguity vs. K-means–GMM agreement). The cross-tabulation is shown in Table 5.

	GMM confident (max_prob ≥ 0.70)	**GMM ambiguous** (< 0.70)
K-means & GMM agree	203 (61.3%)	40 (12.1%)
K-means & GMM disagree	67 (20.2%)	21 (6.3%)

Table 5: Cross-tabulation of GMM self-confidence and K-means–GMM agreement ($n = 331$ LADs). Cells in the bottom row mark methodological disagreement; the bottom-right cell (21 LADs) is doubly ambiguous — GMM is unsure of its own answer *and* disagrees with K-means.

Encoding in the GMM scatter (Sheet 2): colour = `cluster_name` (K-means hard label, ColorBrewer Set2 palette — same as Sheet 1); size = `kmeans_assignment_prob` (GMM's confidence in the displayed K-means label); shape = `kmeans_gmm_agree` (filled circle for agree, "×" for disagree). The shape channel surfaces the 88 disagreement LADs as a visually distinct mark, while size grades each point's K-means-aligned confidence. Boundary cases (low size) and methodological disagreements ("×" shape) are both flagged for individual review before automated programme allocation.

The GMM scatter operates in the same PCA coordinate space — the axes retain their PC1/PC2 interpretation — and adds two information channels (size and shape) that directly answer policy-actionable questions: *"Is this area's cluster assignment reliable, and do both algorithms agree on it?"* The requirement for two distinct dimensionality reduction techniques is satisfied by PCA (Tab 3) and UMAP (Tab 6, benchmark analysis).

Loading Bar Chart

The horizontal loading bar chart uses a **diverging colour scale centred at zero** (orange for negative, blue for positive), appropriate because zero is a meaningful midpoint — positive loadings pull towards one cluster type, negative loadings pull towards the other (Ware, 2004). Sorting by `pc1_abs` descending puts the most influential variables at the top, supporting the user's **Compare** query task.

K-means Cluster Selection and Naming

Selecting $k = 4$: Silhouette score is maximised at $k = 2$ (0.563), and the sharpest inertia reduction occurs at $k = 3$. However, GMM model selection by BIC — which penalises model complexity — is minimised at $k = 4$ (BIC = 2463.8), lower than $k = 2$ (2464.4), $k = 3$ (2466.4), and $k = 5$ (2490.4). Inspecting

$k = 5$ reveals that the additional cluster is an uninterpretable mixture drawn from three separate $k = 4$ clusters, indicating over-segmentation rather than a genuinely new sub-type. K-means silhouette and GMM BIC thus provide complementary but convergent support for $k = 4$, and four clusters yield distinct, non-overlapping intervention types.

Naming methodology — z-score based: Cluster names were derived from the *standardised* deviation of each cluster's feature mean from the national mean, not raw percentage-point deviation. For each cluster c and feature f:

$$z_{c,f} = \frac{\bar{x}_{c,f} - \mu_f}{\sigma_f}$$

where $\bar{x}_{c,f}$ is the mean of feature f over LADs in cluster c, and μ_f, σ_f are the national mean and standard deviation across all 331 LADs. This standardisation enables comparison across features with different natural spreads. A +1.78pp deviation in *bad health* (national $\sigma = 1.35$pp, $z = +1.32$) is statistically more extreme than a +25.96pp deviation in *migrant percentage* (national $\sigma = 11.57$pp, $z = +2.24$ — but on a much wider distribution). Without standardisation, features with naturally wide distributions dominate naming purely by virtue of their spread.

Significance thresholds: Following the standard normal interpretation, $|z| \geq 2$ marks an *extreme* signature (top/bottom 2.5% of LADs); $|z| \geq 1$ marks a *strong* signature (top/bottom 16%); and $|z| < 0.5$ indicates near-average behaviour and is excluded from cluster naming. Applying these thresholds reveals that two of the four clusters carry no $|z| \geq 1$ feature — naming them as "problem" clusters would misrepresent the data. The final names therefore differentiate *action priority*: Cluster 2 and Cluster 3 trigger intervention; Cluster 0 and Cluster 1 require only monitoring.

Feature	**C0** (n=96)	**C1** (n=109)	**C2** (n=41)	**C3** (n=85)
Born outside UK (%)	+0.11	−0.62	**+2.24**	−0.42
Arrived 2011–21 (%)	+0.06	−0.62	**+2.22**	−0.34
No English in HH (%)	−0.06	−0.59	**+2.25**	−0.26
Social rented (%)	−0.23	−0.57	**+1.26**	+0.38
Private rented (%)	−0.23	−0.51	**+1.96**	−0.03
Economically inactive (%)	−0.81	+0.38	**−1.08**	+0.94
Bad/very bad health (%)	−0.90	−0.02	−0.57	**+1.32**
Disabled (%)	−0.80	+0.27	**−1.25**	**+1.16**

Table 6: Cluster z-score profile: standardised deviation of each cluster's feature mean from the national mean. Bold values mark $|z| \geq 1$ (strong signature).

Cluster	n	**max** $\lvert z \rvert$	$\lvert z \rvert \geq 1$ **count**	**Action priority**
C0	96	0.90	0	None (monitoring only)
C1	109	0.62	0	Low (watch only)
C2	41	**2.25**	**7**	**Top (1st priority)**
C3	85	1.32	2	High (2nd priority)

Table 7: Cluster signal intensity summary: maximum absolute z-score and count of features exceeding $|z| \geq 1$, with derived action priority.

Cluster Name	n LADs	Z-score signature & policy meaning
Healthy & Stable (C0)	96	No $\|z\| \geq 1$. Health and welfare burden mildly below average (bad_health $z = -0.90$, econ_inactive $z = -0.81$, disability $z = -0.80$). No intervention required.
Watch Only (C1)	109	No $\|z\| \geq 1$. Mild low-diversity signature (migrant, recent migrant, no English all $z \approx -0.6$); weak economic inactivity rise ($z = +0.38$, below significance threshold). Standard provincial economic support; no migration-specific action.
Migration Pressure (C2)	41	Three *extreme* migration signals ($\|z\| > 2$: born outside UK, recent arrivals, no English) plus strong private rent ($z = +1.96$) and social rent ($z = +1.26$). Highest action priority: integration programmes, ESOL provision, housing supply.
Health Crisis (C3)	85	Strong health and disability burden (bad_health $z = +1.32$, disability $z = +1.16$, econ_inactive $z = +0.94$). Earlier "Housing Insecurity" label rejected: social rent $z = +0.38$ falls below significance. NHS and disability-welfare priority.

Table 8: Final K-means cluster names ($k = 4$, $n = 331$ LADs). Names reflect *action priority* based on z-score signal strength rather than uniform "problem" framing.

Iteration note: Earlier naming attempts (*Moderate Precarity, Economic Exclusion, Language Barrier, Housing Insecurity*; later *Active Economy, Structural Deprivation*) used either un-standardised centroid coordinates or raw percentage-point deviation. Both undervalue features with naturally narrow distributions (e.g., bad health, $\sigma = 1.35$pp) and overvalue features with wide distributions (e.g., migrant, $\sigma = 11.57$pp). The z-score formulation corrects this and reveals that the old "Housing Insecurity" label (social rent $z = +0.38$) and "Economic Exclusion" label (econ inactive $z = +0.38$) lacked statistical grounding — the true differentiators were health/disability burden (C3) and low diversity (C1) respectively.

4.5 Tab 5 — Bayesian Forecast with Trajectory: Encoding Predictive Uncertainty and Temporal Change

Tab 2 defines insecurity as high social renting *and* high economic inactivity. Tab 5 therefore forecasts both dimensions using the same BayesianRidge pipeline, ensuring analytical consistency. The trajectory line charts (Sheets 5-1 and 5-1-b) display the 2011→2021→2031 arc for each LAD as connected solid (actual) and dashed (forecast) line segments, embedding both the descriptive temporal view and the predictive forecast within a single idiom.

Error bars (95% credible interval): Uncertainty is encoded as a first-class visual attribute. Hullman et al. (2015) demonstrate that communicating uncertainty through visual extent (error bar length) leads to better-tuned decisions than presenting point estimates alone.

Five-tier colour scheme: Risk tiers (Critical / High Risk / Moderate Risk / Low Risk / Stable) are derived from empirical percentiles of the forecast distribution rather than a single absolute threshold, ensuring meaningful grouping into levels across all 331 LADs. The 35% absolute threshold is retained as a reference line for policy context. Tiers map directly to policy actions, providing a perceptual anchor (Tversky and Kahneman, 1974) that directs the user's eye to the most urgent cases.

Trajectory line charts (Sheets 5-1, 5-1-b): A connected line chart with year (2011, 2021, 2031) on the X axis and social rent (Sheet 5-1) or economic inactivity (Sheet 5-1-b) on the Y axis displays one line per LAD. The 2011→2021 segment is solid (census actuals); the 2021→2031 segment is dashed (Bayesian forecast) with a shaded 95% CI band. The connected line idiom collapses the entire trajectory — including ranking by magnitude of change — into a single comparison frame, while making the forecast's grounding in observed trend explicit rather than implicit.

Bayesian method justification: BayesianRidge (Pedregosa et al., 2011) was chosen over Ordinary Least Squares (OLS) because only two time points are available (2011, 2021), making OLS overfit perfectly with zero residual and no uncertainty estimate. BayesianRidge's Gaussian prior controls overfitting and produces a posterior predictive distribution, from which the 95% credible interval is computed as mean $\pm 1.96\times$ combined_std (posterior_std $\oplus$ residual_std). The R^2 on training data was 0.987 for social rent and approximately 0.91 for economic inactivity.

Single-axis vs. composite risk: The forecast classifies risk independently on the housing axis (`housing_risk_tier`) and the economic axis (`economic_risk_tier`), then combines the two percentile ranks into a composite tier (`composite_risk_tier`). This separation matters: three inner-London

boroughs — **Hackney**, **Islington**, and **Southwark** — are classified as housing-Critical (forecast SR $\approx 37\%$ by 2031, well above the 35% reference threshold) but only *Moderate Risk* on the composite tier because their forecast economic inactivity is Stable. Conversely, the seven LADs at *composite-Critical* (Hartlepool, Middlesbrough, Newcastle upon Tyne, South Tyneside, Sunderland, Liverpool, Nottingham — see Table 14) are forecast worst on both axes simultaneously, representing a structurally different policy challenge from the inner-London social housing pressure pattern. The dashboard exposes both signals so that the user can choose which lens (single-axis or composite) is relevant to the policy question.

4.6 Tab 6 — UMAP Scatter: Encoding Local Similarity for Outlier Detection

Why UMAP over PCA for Tab 6: UMAP (McInnes et al., 2018) constructs a fuzzy topological graph of the data before embedding, preserving **local neighbourhood structure** more faithfully than PCA for non-linear data shapes. For the peer benchmarking task — finding areas that are genuinely similar across all 8 dimensions but have better outcomes — UMAP proximity is a more reliable similarity measure than PCA proximity (which only preserves linear, variance-maximising structure).

This is the correct trade-off given the task: Tab 3 uses PCA because interpretable axes are needed; Tab 6 uses UMAP because faithful local similarity is needed. Both tabs use the same K-means cluster labels and the same colour palette, ensuring consistency across the dashboard.

Outlier definition (dual-axis with composite Dual rule): For each LAD we compute within-cluster z-scores on the two topic axes: $z_{sr} = (\text{social_rent} - \mu_{sr})/\sigma_{sr}$ and $z_{ei} = (\text{econ_inactive} - \mu_{ei})/\sigma_{ei}$, where μ, σ are the LAD's own cluster mean/std (peers share structurally similar baseline conditions because cluster assignment uses all 8 features). Negative z = LAD performs better than its peers on that axis. The 4-level `outlier_type` encoding is then:

outlier_type	Rule	Shape	Count
`Dual`	$z_{sr} + z_{ei} < -2$ AND both axes negative — composite outperformance	★	15
`Housing only`	$z_{sr} < -1.5$ AND not Dual — strong single-axis deviation on social rent	♦	12
`Economic only`	$z_{ei} < -1.5$ AND not Dual — strong single-axis deviation on economic inactivity	▲	14
`Peer-aligned`	Otherwise — within peer normal range, no flag	•	290

Table 9: Tab 6 outlier classification and 2021 LAD counts ($n = 331$).

Why the composite Dual rule? Applying the strict 1.5σ bar to each axis independently produced zero Dual cases — the two topic dimensions rarely deviate strongly together, so requiring both to clear the bar simultaneously is too restrictive. Loosening to 1.0σ on each axis would weaken the statistical threshold to roughly the 16th percentile, surfacing too much noise. The composite rule rewards LADs whose *overall* performance is clearly better than peers (e.g. -1.0σ on each axis sums to a -2σ joint deviation), so the ★ encoding identifies genuine policy-learning candidates including well-known high-performers (Wokingham, Wandsworth, Merton, City of London, Ribble Valley). The single-axis flags `is_housing_outlier` and `is_economic_outlier` retain the strict 1.5σ bar so they remain defensible as standalone signals.

Shape encoding rationale: Shape is noticed automatically — the four outlier types "pop out" without requiring active visual search (Treisman and Gelade, 1980). Size is varied to amplify the visual hierarchy: Dual ★ uses the largest mark (size 240), the two single-axis types use medium marks (size 130), and Peer-aligned LADs sink to the background (size 35) so the outlier LADs dominate visual attention. Colour remains reserved for cluster identity (four-cluster categorical palette inherited from Tab 3), which keeps shape and colour orthogonal and enables the user to read both channels simultaneously.

Ranked list (Sheet 4): An alternative entry point to the UMAP scatter — all positive outlier LADs (outlier_type $\neq$ Peer-aligned, $n = 41$) sorted by a single composite score, `Outlier Strength`. The score is computed in the Python pipeline as:

$$z_{sr,i} = \frac{\texttt{social_rent}_i - \bar{x}_{sr}^{\text{cluster}}}{\sigma_{sr}^{\text{cluster}}}, \qquad z_{ei,i} = \frac{\texttt{econ_inactive}_i - \bar{x}_{ei}^{\text{cluster}}}{\sigma_{ei}^{\text{cluster}}}, \qquad z_i^{\text{combined}} = z_{sr,i} + z_{ei,i},$$

where $\bar{x}^{\text{cluster}}$ and σ^{cluster} are the mean and standard deviation over *all* LADs in cluster $c(i)$ (not Peer-aligned only). In Tableau, `Outlier Strength` $= -z^{\text{combined}}$ (sign-flipped so that the most negative

combined z — the strongest outperformer — sorts to the top of the list).

This z-quantity differs from the profile heatmap baseline (Sheet 3): the heatmap uses Peer-aligned-only median/std for robust centering, whereas the ranked-list z uses cluster-wide mean/std because its purpose is *classification and ranking* rather than profile description. The same LAD's z value will therefore differ numerically between the two sheets — this is intentional and reflects the different analytical question each sheet answers.

Crosstab (Sheet 2): The comparison table (selected LAD vs. cluster median vs. national median) uses LOD (Level of Detail) expressions rather than `WINDOW_MEDIAN` to compute cluster and national statistics correctly regardless of view filter.

Outlier-type profile heatmap (Sheet 3): A group-level methodology validation view: rows = the four `outlier_type` categories, columns = 8 features, each cell = the *mean of per-LAD z-scores* within that type. The computation proceeds in two explicit stages.

Stage 1 — per-LAD z-score (cluster-aware). For each LAD i and feature f, the baseline is the *Peer-aligned LADs within i's own cluster* only:

$$z_{i,f} \;=\; \frac{x_{i,f} \;-\; \widetilde{x}^{\,\text{peer}}_{c(i),\,f}}{\sigma^{\text{peer}}_{c(i),\,f}},$$

where $c(i)$ is LAD i's cluster assignment (inherited from Tab 3), $\widetilde{x}^{\,\text{peer}}_{c(i),\,f}$ is the *median* of feature f over the Peer-aligned LADs in cluster $c(i)$, and $\sigma^{\text{peer}}_{c(i),\,f}$ is their standard deviation. Median is preferred over mean for centering because several features — most notably `migrant_pct` and `no_english_pct` — are right-skewed within clusters: a small number of high-concentration LADs would pull the mean above the typical Peer-aligned value and understate deviation for most LADs. The median places the baseline at the actual central Peer-aligned observation, which is the appropriate comparison point for "how different is this outlier from a typical peer?" Using Peer-aligned LADs only as the baseline avoids self-bias: if the full cluster population were used, the outlier LADs themselves would shift the centre toward them, shrinking the apparent deviation and weakening the secondary-feature signal the heatmap is designed to surface.

Stage 2 — cross-cluster group aggregation. The cell value for outlier type t and feature f pools per-LAD z-scores across all four clusters:

$$\text{cell}_{t,\,f} \;=\; \frac{1}{|t|}\sum_{i \in t} z_{i,\,f}, \qquad t \in \{\text{Dual, Housing only, Economic only, Peer-aligned}\}.$$

This pooling is a practical choice forced by small within-cluster counts (Table 10): three cluster × type cells contain only 1 LAD each (Health Crisis × Dual; Migration Pressure × Housing only; Migration Pressure × Economic only). A cluster-by-cluster heatmap would reduce those cells to a single observation, which is not a group profile. Cross-cluster pooling is methodologically defensible because the Stage 1 z-scores are already cluster-normalised — each deviation is measured in units of its own peer group's spread, so pooling combines comparable quantities rather than raw percentages.

	Dual	**Economic only**	**Housing only**	**Peer-aligned**
Health Crisis	1	3	3	78
Healthy & Stable	3	6	3	84
Migration Pressure	3	1	1	36
Watch Only	8	4	5	92

Table 10: LAD counts by cluster and outlier type ($n = 331$). Cells with $n = 1$ make a cluster-by-cluster heatmap statistically unworkable and justify cross-cluster pooling in Stage 2 of the profile computation.

Heatmap findings. The cell values in Figure **??** validate the outlier classification and surface three policy-relevant patterns beyond the definitional axes.

Dual ($n = 15$). All eight features are negative, including secondary indicators: bad_health (-1.52σ) and disability (-1.53σ) are the strongest deviations alongside the definitional econ_inactive (-1.40σ) and social_rent (-1.32σ). This confirms the classification: Dual LADs are not accidentally low on two metrics but are structurally well-integrated communities across the full feature space. The migration features are near zero (-0.02 to -0.11σ), confirming that cluster membership already controls for migrant concentration — the outperformance is on welfare and housing outcomes, not on having fewer migrants.

Housing only ($n = 12$). Social rent reaches -2.01σ (definitional), but economic inactivity is *positive* at $+0.84\sigma$ — these LADs perform worse than their cluster peers on economic integration despite achieving housing efficiency. This trade-off limits the scope of policy transfer: only housing-specific interventions (planning policy, social housing management) are candidates for replication; economic integration practices from these LADs should not be adopted.

Economic only ($n = 14$). Economic inactivity is -1.94σ (definitional), but social rent is $+0.90\sigma$ — the mirror trade-off. Additionally, migrant ($+0.85\sigma$), recent_migrant ($+0.92\sigma$), and no_english ($+0.96\sigma$) are all positive, indicating that Economic only outlier LADs are high-migration LADs that achieve economic integration under housing pressure. Policy transfer scope is the inverse: economic integration programmes (ESOL, employment linking) are candidates for replication; housing approaches are not.

Three interpretive notes on the heatmap mechanics follow. *(i) Peer-aligned row.* When Peer-aligned LADs are compared against their own cluster's Peer-aligned baseline (Stage 1), the mean z per feature should be zero. The small non-zero values observed in the heatmap (range -0.1 to $+0.3\sigma$) arise because the centering uses median rather than mean and because unequal cluster sizes introduce a weighting asymmetry in Stage 2. These near-zero values confirm that the Peer-aligned category is correctly tuned as the reference level. *(ii) Two distinct z-quantities.* The 1.5σ outlier classification threshold applied in the UMAP scatter (Sheet 1) uses cluster-wide μ/σ (all 331 LADs), whereas the heatmap baseline uses Peer-aligned-only statistics (290 LADs, excluding outlier LADs). The two z-quantities are not numerically identical and should not be conflated. *(iii) Group mean, not per-LAD.* A -1.32σ in the Dual × social_rent cell is the average across all 15 Dual LADs; individual LADs may deviate much more or less strongly. Per-LAD drill-down is provided through the Comparison Crosstab (Sheet 2). The heatmap's role is a *methodology check* — validating that the four outlier types form consistent profiles on features *beyond* the definitional axes (z_{sr}, z_{ei}) — not a decision artefact. Diverging red/white/blue palette centred at zero, clipped at $\pm 2\sigma$.

5 Evaluation

5.1 Evaluation Framework

Following Munzner (2014) (Ch. 4), visualisation quality is assessed at four nested levels:

Level	Threat	Validation approach
Domain Situation	Wrong problem	Target user needs derived from public policy literature (Home Office, DLUHC operational context); confirmed against UK integration policy frameworks
Task & Data Abstraction	Wrong abstraction	Discussion group analytic task — see §5.2; downstream informal evidence of tool utility
Visual Encoding & Interaction	Wrong idiom	Justified against perceptual and cognitive principles in Section 4; immediate review against known guidelines
Algorithm	Slow algorithm	PCA: $O(\min(n,p)^3)$ trivially fast at $n = 331$; UMAP: $O(n \log n)$; BayesianRidge: $O(n^2)$; all ran in under 30 seconds

Table 11: Munzner's four-level validation framework applied to this dashboard.

The primary validation effort is at the **idiom level** and the **abstraction level**, addressed through a discussion group task study described below.

5.2 Evaluation Design

Scope and Rationale

The evaluation was conducted during a scheduled lab session on 29 April 2026. Three fellow students participated as evaluators within the 15-minute per-student session format specified by the course. The scope was deliberately limited to **Tab 1** and **Tab 2**, which together implement the complete primary decision pathway (Decision 1: Geographic Screening). Tab 1 establishes the geographic distribution of migrant concentration; Tab 2 classifies areas by compound housing and economic insecurity. These two tabs were chosen because they constitute the entry point for every subsequent policy decision: no LAD

reaches the shortlist for Tabs 3–6 without first passing the Tab 1 + Tab 2 dual filter. The evaluation instrument was distributed via a Google Form.[1]

Task Scenario

Participants were given the following structured task:

> *(Task A — Geographic Screening, Tab 1): Using the map and bar chart on Tab 1, identify the **3 Local Authority Districts in the North West** with the highest migrant population share (%). Write their names.*
>
> *(Task B — Intervention Type, Tab 2): For each of the 3 areas above, which quadrant do they fall into on Tab 2? Options: "1: High social rent & High economic inactivity"; "2: High social rent only"; "3: High economic inactivity only"; "4: Below average both". Write: [Area name]: [Quadrant Number] for each.*
>
> *(Task C — Reference Line Comprehension, Tab 2): What do the two dashed reference lines on Tab 2 represent? How many LADs appear in the "High social rent & High economic inactivity" quadrant (top-right)?*

Ground truth answers for accuracy scoring:

- **Task A**: Top-3 North West LADs by `migrant_pct_2021` readable from the bar chart *after applying the region filter*: Oldham, Manchester, Salford (accept any top-3 visible from the North West subset).
- **Task B**: All three ground-truth North West LADs fall in Q1 (above national average on both social rent and economic inactivity).
- **Task C**: Reference lines = national average social renting rate (15.9%) and national average economic inactivity rate (39.5%); 64 LADs in Q1.

Measurement Protocol

Qualitative measures — post-task questionnaire:

Q	Question	Scale
Q1	How easy was it to identify the 3 highest-risk North West LADs using the map and bar chart on Tab 1?	1 (very difficult) – 5 (very easy)
Q2	Did the four quadrant labels on Tab 2 make sense without extra explanation? (Priority 1 / High social rent / High inactivity / Below average)	Yes / Partially / No
Q3	Did the national average reference lines on Tab 2 help you classify each area into a quadrant?	Yes / Somewhat / No
Q4	Was anything on Tab 1 or Tab 2 confusing or misleading?	Open text
Q5	What one thing would you change or add to make either tab easier to use?	Open text

Table 12: Post-task questionnaire items (administered via Google Form, `https://forms.gle/25VEzrwNE9T7Qghr9`).

5.3 Observations and Measurements

Participants: $N = 3$ discussion group members, all fellow students with data literacy but no prior knowledge of UK housing policy or the specific dashboard. All three were non-native English speakers, which is noted as a contextual factor for interpreting the Task A outcome below.

Task A — Geographic Screening (Tab 1):

- **Responses**: All three participants identified Brent, Westminster, and Kensington & Chelsea — three London boroughs.
- **Accuracy**: 0/3 against ground truth (correct: Oldham, Manchester, Salford from the North West region after applying the region filter).

[1] Survey link: `https://forms.gle/25VEzrwNE9T7Qghr9`

- **Observation**: Participants correctly applied the chart idiom — they read the national Top-20 bar chart and identified the highest bars by length. However, none applied the North West region filter before reading the chart. The task required a filter interaction step that no participant initiated, indicating that the region dropdown was not perceived as a necessary first action. The error is therefore diagnostic of a *filter prominence* issue at the idiom level, not a misunderstanding of the underlying data representation.
- **Q1 mean score**: 4.67 / 5 (individual scores: 5, 5, 4). The high rating reflects participants' success in applying the global ranking idiom; it does not reflect performance on the regionally scoped task.

Task B — Quadrant Classification (Tab 2):

- **Responses**: All three classified Brent and Westminster as Quadrant 1 ("High social rent & High economic inactivity"); the third area was left blank or marked N/A by all participants.
- **Accuracy**: Technically correct for the LADs the participants identified — Brent and Westminster do fall in Q1 of the scatter. The incomplete third response reflects ambiguity for borderline points near the reference lines.
- **Observation**: Participants located familiar London borough marks on the scatter and read off their quadrant membership using the reference lines. The Q1 label and reference line combination guided classification without additional instruction.
- **Q2 responses**: Partially (3/3). All participants found the quadrant labels partially interpretable without explanation.

Task C — Reference Line Comprehension (Tab 2):

- **Responses**: All three correctly identified the reference lines as national average values ("average percentage"; "Avg. social Rent pct 2021"; "Avg. social percentage 2021"). No participant reported the specific threshold values (15.9% and 39.5%) or the Q1 count of 64 LADs.
- **Accuracy**: 3/3 on reference line conceptual identification; 0/3 on exact count of Q1 LADs (question not answered by any participant).
- **Q3 responses**: Yes (3/3). All participants confirmed that the reference lines helped them classify areas into quadrants.

Qualitative Feedback (Q4 and Q5):
Q4 (anything confusing or misleading):

- "the filter could be better" (2 participants)
- "dashboard 2 can be much more easy to look at" (1 participant)

Q5 (one thing to change or add):

- "add Indicator card OR improve filter function more user-friendly" (1 participant)
- "Improve the filter" (1 participant)
- "Add visible count on the dashboard" (1 participant)

5.4 Validation Assessment

The discussion group study provides evidence at two Munzner validation levels (Munzner, 2014):

1. **Idiom-level downstream validation**:

   Success: The reference line idiom (Tab 2) was validated: Q3 "Yes" from all three participants confirms that the national average reference lines successfully communicated the quadrant boundary without prior instruction. The high Q1 rating (4.67 / 5) validates the choropleth + ranked bar chart combination as legible for geographic ranking tasks without explanation.

Filter interaction failure: The region dropdown (Tab 1) was not applied by any participant. This constitutes an idiom-level failure for the filter interaction mechanism — specifically, the absence of a noticeable visual cue directing users to scope the view by region before reading the bar chart. The current design provides a linked multi-value dropdown in the filter panel (connected across all four Tab 1 sheets via a shared parameter), but no in-chart annotation prompts its use. *Partial mitigation already implemented*: the bar chart encodes region via hue on each bar, allowing users to visually identify North West LADs by colour without applying the dropdown. *Proposed improvement*: add an instructional annotation adjacent to the region dropdown (e.g., "Filter by region to scope all views") to make the filtering step explicit.

2. **Abstraction-level downstream validation**:

   Participants correctly classified Brent and Westminster into Q1 (Task B), providing informal evidence that the two-dimensional insecurity abstraction (social rent × economic inactivity, divided by national average reference lines) correctly communicates combined risk to target-adjacent users (Munzner, 2014). The "Partially" responses on Q2 indicate that the quadrant label abstraction is interpretable but would benefit from inline threshold values (e.g., "$>$ 15.9% social rent & $>$ 39.5% inactivity") rather than relying on users to infer the meaning of "high" from the reference line position alone.

3. **Identified limitations and design responses**:

   - **Filter prominence (primary finding)**: All three evaluators cited the region filter as a usability concern (Q4 and Q5). The current design provides a linked multi-value dropdown but does not guide users to apply it as a first step. Proposed improvement: add a prompt annotation or a default "Select a region" placeholder adjacent to the filter to make the action self-evident.
   - **Quadrant label clarity**: The "Partially" response (Q2, 3/3) indicates the labels are partially self-explanatory. Proposed improvement: embed the national average threshold values directly in the quadrant annotation text to eliminate the inference step about what "high" means.
   - **Count indicator**: One participant (Q5) requested a visible LAD count per quadrant. The current design does not display this summary statistic as a fixed annotation; the count of 64 Q1 LADs is derivable from hovering individual marks but is not shown as a persistent label. Proposed improvement: add a fixed text annotation within the upper-right quadrant (e.g., "64 of 331 LADs") to provide an immediate numerical anchor.

Limitations of this evaluation: The three participants were non-native English speakers with data literacy but no domain expertise in UK housing policy. The Task A error (London LADs instead of North West) may partly reflect unfamiliarity with the "North West" as a specific ONS region name rather than a pure visualisation usability failure. As Andrews (2008, cited in Munzner 2014, p. 79) notes, early-stage methods lead to better and more usable systems but do not offer validation of an approach or provide evidence of the superiority of an approach for a particular context. The results are therefore treated as early-stage evidence informing design improvements, not as proof of the dashboard's superiority over alternatives.

6 Decision Scenario Walkthrough

To demonstrate that the dashboard's funnel structure (§3, Table 4) supports an end-to-end policy decision rather than five disconnected analyses, this section reproduces a complete walkthrough conducted on the built dashboard. Each step exercises one decision scenario and passes a concrete LAD pool to the next; the final step yields the policy-action artefact: a peer LAD plus 1–2 features for a case-study brief.

6.1 Decision 1-1 (Tab 1) — Migrant-Pressure Screening

Reading Sheets 1-1 (symbol map, 2021) → 1-2 (rank by share) → 1-3 (where the share grew) → 1-3-b (Q1: Δshare > 0 *and* Δcount > 0), with the region filter scoped to London and the South East, surfaces six LADs in the Q1 prior-target quadrant (Table 13). Reading these LADs' `cluster_name` from Tab 3

(Decision 3 reads downstream) reveals that all six belong to the *Migration Pressure* cluster — internal validation that Tab 1's geographic narrowing produces a cluster-consistent pool.

LAD	Region	Δshare (pp)	Δcount	Tab-2 quadrant	Tab-3 cluster
Harrow	London	+6.21	+26,542	Q3 (low SR + low EI)	Migration Pressure
Hounslow	London	+6.36	+32,707	Q4 (high SR + low EI)	Migration Pressure
Tower Hamlets	London	+3.73	+35,791	Q4 (high SR + low EI)	Migration Pressure
Reading	South East	+8.68	+19,711	Q4 (high SR + low EI)	Migration Pressure
Slough	South East	+4.99	+15,042	Q4 (high SR + low EI)	Migration Pressure
Oxford	South East	+6.50	+13,414	Q1 (high SR + high EI)	Migration Pressure

Table 13: Walkthrough of Decision 1-1 / 1-2 / 3 on London and South East regions. The Q1 prior-target pool of Tab 1 (1-3-b) yields six LADs. After cross-tab joins, all six map onto the same Tab-3 cluster (Migration Pressure), with one LAD (Oxford) additionally qualifying as a Tab-2 Q1 emergency case (above both SR 15.85% and EI 39.51% thresholds).

6.2 Decision 2 (Tab 5) — Forward-Looking Forecast

Filtering Sheet 5-2 by `composite_risk_tier = Critical` surfaces seven LADs (Table 14). The non-obvious finding is that **none of the seven Critical LADs are in the Migration Pressure cluster from Tab 1's pool**: all seven sit in the *Health Crisis* cluster. The funnel therefore separates *which areas have visibly risen migrant share now* (Migration Pressure, surfaced by Tabs 1–2) from *which areas will be worst-off in 2031 on combined housing and economic forecasts* (Health Crisis, surfaced by Tab 5). Treating either signal alone would systematically miss the other priority.

LAD	Region	SR 2031 (%)	SR 2031 CI lower	EI 2031 (%)	Composite score	Cluster
Newcastle upon Tyne	North East	25.80	24.49	54.34	92.6	Health Crisis
South Tyneside	North East	27.90	26.58	53.55	91.6	Health Crisis
Sunderland	North East	25.25	23.93	53.73	91.0	Health Crisis
Hartlepool	North East	22.37	21.05	54.86	90.8	Health Crisis
Middlesbrough	North East	22.19	20.88	54.80	90.3	Health Crisis
Liverpool	North West	25.02	23.71	53.63	90.5	Health Crisis
Nottingham	East Midlands	24.21	22.90	55.34	94.2	Health Crisis

Table 14: Walkthrough of Decision 2 on Tab 5. The seven LADs at `composite_risk_tier = Critical` for 2031 are concentrated geographically (North East + North West + East Midlands) and clustered exclusively in the Health Crisis classification. Source: `tab5_forecast.csv`.

6.3 Decision 4 (Tab 6) — Peer Benchmark Recommendation

For each of the two priority clusters surfaced by the funnel (Migration Pressure from Tabs 1–2; Health Crisis from Tab 5), Tab 6 ranks within-cluster Dual benchmarks ($z^{\text{combined}} < -2$, both axes negative). Sheet 6-3 then breaks down the per-feature gap between the benchmark LAD and its cluster median, identifying the 1–2 indicators on which a policy-transfer case study should focus.

Cluster	Benchmark LAD	z_{sr}	z_{ei}	z^{combined}	Largest negative gaps vs. cluster median (Sheet 6-3)
Migration Pressure	Wandsworth	−0.47	−2.10	−2.57	Econ inactive −9.06pp; no English / disability −2.20pp; migrant −1.96pp; bad health −1.24pp
Migration Pressure	City of London	−1.01	−1.44	−2.45	Social rent −6.62pp; econ inactive −6.37pp; bad health −2.89pp; no English −1.66pp
Migration Pressure	Merton	−1.09	−0.91	−2.00	Social rent −7.28pp; econ inactive −4.25pp; recent migrant −0.75pp
Health Crisis	Barrow-in-Furness	−1.31	−0.79	−2.11	Social rent −7.40pp; migrant −3.60pp; private rent −1.78pp; econ inactive −1.72pp

Table 15: Walkthrough of Decision 4 on Tab 6. *Top* three rows: Dual benchmarks inside the Migration Pressure cluster (3 of 41 LADs). *Bottom* row: Barrow-in-Furness, the only Dual benchmark in the entire 85-LAD Health Crisis cluster — a singularity that is itself a policy finding (positive outlier behaviour is rare in Health Crisis, so the case-study transfer pool is shallow). Aggregate diagnostic across all 4 clusters: the Dual outlier type ($n = 15$) shows mean z-deviations of disability −1.53, bad health −1.52, econ inactive −1.40, social rent −1.32 relative to its peer-aligned baseline (Sheet 6-2, heatmap), confirming that Dual outlier LADs outperform their peers across the full feature space, not merely on the two definitional axes.

6.4 Final Policy Action

Combining the three walkthrough tables yields concrete, checkable policy-action briefs of the form:

> *For Newcastle upon Tyne (Health Crisis cluster, Critical composite risk by 2031, forecast SR* 25.8% *and EI* 54.3%*), commission a policy-transfer case study on Barrow-in-Furness (same cluster,* $z^{\text{combined}} = -2.11$*). Focus the case study on Barrow's social-rent management (gap* −7.4*pp vs. Health Crisis cluster median) and economic-integration approach (econ inactive gap* −1.7*pp). The Health Crisis cluster contains only one Dual benchmark, so this transfer candidate is unique within the cluster — the policy-evaluation team should treat the recommendation as exploratory rather than confirmatory.*

The same template applies to Migration Pressure priority LADs (e.g. Tower Hamlets, Oxford), drawing on Wandsworth (economic-integration interventions: ESOL, employment linking, in-work support) or City of London / Merton (social-rent management) depending on which feature gap dominates the priority LAD's profile.

7 Conclusion

7.1 What I Have Learned About the Socio-Economic Problem

This analysis reveals several policy-relevant findings not apparent from any single indicator:

1. **Migration and deprivation are orthogonal, not correlated**: PCA reveals that PC1 (New Arrival & Language Barrier) and PC2 (Welfare Dependency) are largely independent axes. Many high-migrant LADs (London boroughs) have *better* economic activity rates than low-migrant LADs with old industrial histories (e.g., County Durham, Merthyr Tydfil). The stereotype that migrant-dense areas are economically deprived is not supported by the data at LAD level.

2. **Today's migrant-pressure pool and tomorrow's risk pool do not coincide**: The walkthrough in §6 surfaced two non-overlapping priority sets. The six Tab-1 / Tab-2 prior-target LADs in London and the South East (Harrow, Hounslow, Tower Hamlets, Reading, Slough, Oxford) are all in the Migration Pressure cluster, while the seven LADs at *Critical* composite risk for 2031 (Newcastle upon Tyne, South Tyneside, Sunderland, Hartlepool, Middlesbrough, Liverpool, Nottingham) all sit in the Health Crisis cluster. Treating either signal alone — "where are migrants now?" or "where will risk peak in 2031?" — would systematically miss the other priority. The funnel is what makes the divergence visible.

3. **Cluster classification reveals divergent intervention needs**: The 41 Migration Pressure LADs (three migration features at $|z| > 2$) need ESOL investment and integration programmes; the 85 Health Crisis LADs (bad health $z = +1.32$, disability $z = +1.16$) need NHS and disability-welfare resourcing. These two action-priority clusters overlap only partially in geographic distribution and not at all in required intervention, confirming that aggregate statistics mask strategically different local situations.

4. **Positive-outlier candidates are unevenly distributed across clusters**: The Migration Pressure cluster (41 LADs) contains three Dual benchmarks (Wandsworth, City of London, Merton — all London) suitable for case-study transfer; the Health Crisis cluster (85 LADs) contains only one (Barrow-in-Furness). This asymmetry has direct policy-design implications: case-study transfer is a credible strategy for Migration Pressure LADs but a constrained one for Health Crisis LADs, where the evidence base for within-cluster "what works" is shallow and additional national or cross-cluster benchmarks may be required.

7.2 What I Have Learned About Information Visualisation

The most important insight from this coursework is that **the task abstraction stage is harder and more important than the idiom design stage**. Choosing which variables to put on which axes — for example, recognising that within-cluster z-score deviation (not raw percentage gap) is the right basis for cluster naming because it controls for variables with naturally wide distributions — required understanding the underlying data at a depth that precedes any visualisation design decision. Munzner's (2014) observation that "the abstraction stage is often the hardest to get right" is directly confirmed by this experience.

The difference between PCA and UMAP, and why each is appropriate for a different tab, taught me that dimensionality reduction is not a technique to be applied mechanically — it is a design decision that encodes an implicit claim about what "similarity" means. PCA says: similarity along the directions of highest variance. UMAP says: similarity in local neighbourhood topology. For a policy user who needs to understand *why* areas are similar (Tab 3), PCA is the correct claim. For a policy user who needs to find *genuinely similar* peers (Tab 6), UMAP is the correct claim.

Finally, the evaluation exercise confirmed that **uncertainty visualisation changes decision behaviour**. Participants who saw credible intervals on the forecast distinguished between high-confidence and uncertain cases in ways that a point-estimate-only visualisation would not support. This is the core value proposition of Bayesian methods in a policy dashboard context: not just predicting what will happen, but communicating how confident the prediction is.

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